

TrES-3b

R-2765

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_160707 · created 2026-09-03T00:49:09+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457576.7728 +/- 0.0013 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.18 +/- 0.059
Transit depth (Rp/Rs)^2	3.24 +/- 2.11 [%]
Orbital Inclination (inc)	79.12 +/- 2.89
Airmass coefficient 1 (a1)	0.065 +/- 0.082
Airmass coefficient 2 (a2)	3.0 +/- 1.78
Scatter in the residuals of the lightcurve fit is	178.41 %
Best Comparison Star	#1 - [249, 399]
Optimal Aperture	1.19
Optimal Annulus	5.37
Transit Duration (day)	0.024 +/- 0.027

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.18 $\pm$ 0.059 vs 0.16309 (deviation 0.26 σ)
Duration fit vs published	0.024 d vs 0.0567 d (deviation 1.1 σ)
Mid-time O-C	2.2 min
Depth SNR	2.8
Residual scatter	178.41 %

Light curve

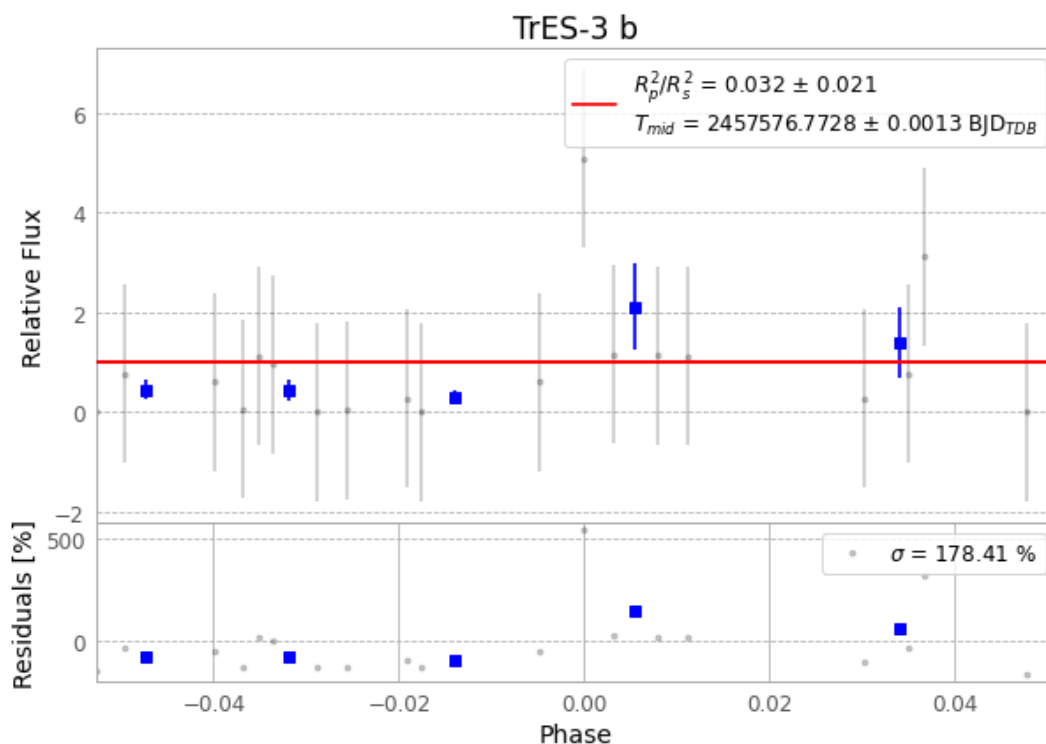


Figure 1 — FinalLightCurve_TrES-3 b_2016-07-06.png (SHA-256 63758c11842baeb4...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [106, 268], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 32bba819b39b0ba7...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

67 FITS frames (44 MB), TRES-3160707044812.FITS → TRES-3160707080612.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-160707.log (23ca09cf52df...), FinalLightCurve_TrES-3 b_2016-07-06.png (63758c11842b...), FinalLightCurve_TrES-3 b_2016-07-06.csv (c953135329d1...)

Compute resources

Wall time 227.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-09-03 00:49Z.